Inorganic Cume: Inorganic Synthesis
September 14, 2001

1. Sulfuric acid is the number one manufactured chemical in the world. Give equations for the series of reactions that show how sulfuric acid is prepared, starting with elemental sulfur.

2. Another inorganic chemical of major industrial importance is ammonia. Give the equation for the manufacture of ammonia.

3. Give a general description for the preparation of Buckministerfullerene, C_{60}.

4. Give a laboratory preparation of ozone.

5. Many complexes of Co(III) start with $CoCl_2$. How is the oxidation usually done?

6. Write the equation that accounts for the formation of NO_2 that is part of the smog of cities such as Los Angeles and Denver. Where do this reaction take place?

7. Give the equation for the lab preparation of anhydrous iron (III) chloride.

8. What is the main reducing agent in the industrial preparation of iron from the iron ore Fe_3O_4?

9. Give the equation for a reaction that produces a compound of xenon.

10. For the preparation of S_4N_4, give a reasonable S-containing reagent and a reasonable N-containing reagent.

11. Describe the industrial preparation of aluminum metal from aluminum ore.

12. Give an example of an inorganic synthesis reaction that is best carried out photochemically, and explain why the photochemical reaction is preferred over the reaction involving just the heating of the reagents.

13. For inorganic syntheses involving oxidation or reduction in aqueous solution, give a very strong oxidizing agent and a very strong reducing agent.